Do LLM Judges Penalise the Script?

A script-controlled audit of LLM-as-judge scoring for Hindi in Devanagari versus romanized orthographies

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https://abraar237.github.io/script-bias-llm-judges/

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Abstract

Large language models now grade other language models. These LLM judges feed leaderboards, reward models, and production quality gates, and every such pipeline assumes that the writing system of an answer does not move its score. We test that assumption directly for Hindi, a language written both in its native Devanagari script and, by hundreds of millions of users online, in the Latin alphabet. We hold content fixed and vary only the script: 150 Hindi instruction and response pairs at three quality tiers, each rendered in native Devanagari and three romanizations (scholarly IAST, diacritic-stripped ASCII, and natural user-style Hinglish), scored blind by six judges from three model families. We pre-registered the direction of the effect: we predicted romanized text would score lower. The data reversed our prediction, and we report that reversal plainly. First, Gemini 3.6 Flash scores identical romanized content **2.2 to 3.4 points higher** than Devanagari on a 100-point scale (all $p \leq 0.003$), and the inflation concentrates at **6.5 to 7.8 points** in medium-quality responses, where a judge has the most discretion. Second, the effect shrinks but survives at the frontier tier: Gemini 3.1 Pro inflates by 1.1 points overall (all $p \leq 0.003$). Third, the bias changes shape at the open-weights tier: Qwen2.5-7B inflates scholarly IAST by **7.6** points and ASCII by **6.0** points, concentrated at $+\mathbf{17.8}$ points on low-quality items, yet is flat on natural Hinglish ($+0.1$, $p = 0.97$). Fourth, three Claude judges are script-invariant to within half a point, except a small penalty in our pre-registered direction: Opus 5 on ASCII, -1.2 points ($p = 0.002$). Fifth, the obvious fix fails: instructing the judge to ignore the script leaves the inflation intact. Finally, mechanism probes rule out tokenization length (romanized Hindi costs 1.33 to 1.98 times the tokens of Devanagari, but token inflation does not predict score shift, $|r| \leq 0.18$) while showing the script is perceived: judge rationales name the orthography on 41 to 57 percent of romanized items versus 1 percent of Devanagari items. Script bias in LLM judges is real, heterogeneous across judges in size and sign, and cannot be instructed away.

1 Introduction

Evaluation of language models has largely been handed to other language models. An LLM judge reads a question and an answer and returns a score (Zheng et al., 2023), and that score then does consequential work: it ranks systems on leaderboards, it becomes the reward signal that trains the next model, and it gates which responses production systems are allowed to ship. The reliability literature has catalogued many failure modes of these judges, including position bias (Wang et al., 2024), verbosity and self-enhancement bias (Zheng et al., 2023), self-preference driven by self-recognition (Panickssery et al., 2024), and taxonomies now spanning a dozen or more bias types (Ye et al., 2025; Zhou et al., 2026b). Every catalogue so far shares one silent assumption: that the *writing system* of the judged text does not move the score.

That assumption matters most for languages with two written lives. Hindi is the clearest case. Its official script is Devanagari, but a very large share of Hindi on phones, in chat, and in social media is typed in the Latin alphabet, a practice called romanized Hindi or Hinglish (Sheth et al., 2025). The two forms carry identical content. A judge that scores them differently is not measuring quality; it is measuring orthography.

This paper asks one narrow question with a clean design: if we hold the language and the content of an answer fixed and change only its script, does an LLM judge's score move? Because the content on both paths is identical (Figure 1), nothing else can explain a gap: not answer quality, not language difficulty, not topic. Prior multilingual judge studies could not make this claim because they compared different languages, so language and script were entangled (Hada et al., 2024b; Fu and Liu, 2025; Zhou et al., 2026a). We untangle them.

We pre-registered the direction of our prediction before collecting data: we expected romanized text to score

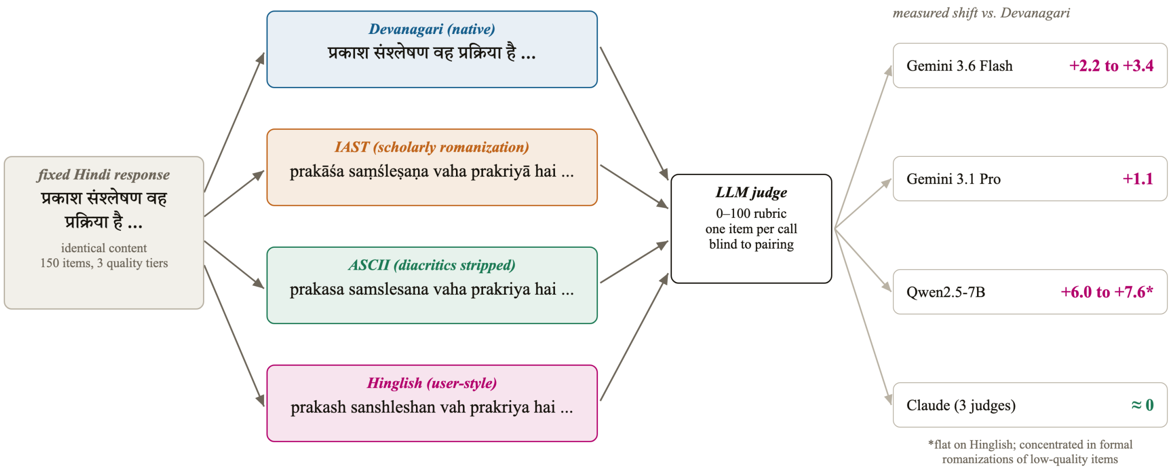

Figure 1: We hold one Hindi response fixed and vary only its writing system, then ask six LLM judges to grade each rendering blind. Content is identical along every path, so any score gap is script bias by construction. The right column previews the measured outcome: both Gemini judges and the open-weights Qwen judge score romanized renderings *higher* than the identical Devanagari original, while three Claude judges are script-invariant. The starred Qwen entry is flat on user-style Hinglish; its inflation concentrates in formal romanizations of low-quality items (Section 4).

lower, by analogy to the task-performance literature, where romanized input degrades model accuracy by up to 24 points (Khullar et al., 2025) and inflates perplexity over 300-fold (Rajapakse and Weerasinghe, 2026). The data reversed the prediction. Romanized Hindi scores *higher* on Gemini and Qwen judges, the inflation concentrates exactly where a judge has discretion, it persists when the judge is told to ignore the script, and it is absent in the Claude family. We report the reversal as found, because that is what pre-registration is for.

Our contributions:

- **Script-controlled judge audit** (§3): the first audit of LLM-as-judge scoring that isolates script as a treatment: 150 fixed Hindi items, four orthographies, six judges from three families, 4,200 scripted judgments.
- **A new judge-bias axis** (§4): statistically significant script bias, opposite in sign to our pre-registered prediction, up to +7.8 points on medium-quality items under Gemini 3.6 Flash and +17.8 on low-quality items under Qwen2.5-7B, absent in Claude judges. Orthography appears in no existing judge-bias taxonomy (Ye et al., 2025; Zhou et al., 2026b).
- **A negative mitigation result** (§4.3): a script-blindness instruction does not reduce the bias, so the practitioner remedy is judge selection and script-stratified reporting, not prompting.
- **Mechanism probes** (§4.4): token-length inflation is ruled out as the driver, and the judges' own rationales show the script is perceived while the miscalibration happens anyway.

2 Related Work

LLM-as-judge and its biases. Zheng et al. (2023) established the LLM-as-judge paradigm with MT-Bench and Chatbot Arena and catalogued position, verbosity, and self-enhancement biases. Wang et al. (2024) showed pairwise judgments can be flipped by candidate order alone. Panickssery et al. (2024) traced self-preference to self-recognition. CALM (Ye et al., 2025) quantified twelve bias types through automated perturbation, and JudgeBias-Bench (Zhou et al., 2026b) extended the taxonomy with debiasing baselines. Orthography is absent from all of these.

Multilingual judging. Hada et al. (2024b) first calibrated GPT-4 judging against native-speaker judgments across eight languages and found systematic overscoring, worst for low-resource languages. METAL (Hada et al., 2024a) and PARIKSHA (Watts et al., 2024) measured judge-human agreement across languages, with PARIKSHA covering ten Indic languages. MM-Eval (Son et al., 2024) and Fu and Liu (2025) stress-tested judges across up to 122 languages and found roughly 0.3 cross-lingual consistency. BabelJudge (KC, 2026) measures judge reliability including Hindi, in native script only, and Zubiaga et al. (2026) study multilingual judge construction on Latin-script languages. Closest to us, Zhou et al. (2026a) audit judges on semantically identical content across 23 languages and find lower-resource languages scored *more* generously, noting a Latin-script advantage observationally. In every one of these studies script co-varies with language; none isolates it.

Script and romanization in LLMs. RomanSetu (Husain et al., 2024) showed romanization can serve as an efficient interface for adapting English-centric models to Indic languages. Transliteration-based adaptation has a longer history (Purkayastha et al., 2023; Xhelili et al., 2024), supported by large transliteration resources (Madhani et al., 2023). On the deployment side, Script Gap (Khullar et al., 2025) measured task-accuracy drops of up to 24 points on romanized versus native-script inputs across five Indian languages, Rajapakse and Weerasinghe (2026) measured over 300-fold perplexity degradation on romanized Sinhala, and Indi-RomCoM (Das et al., 2026) benchmarked degradation on romanized code-mixed instructions. Code-mixed evaluation resources include COMI-LINGUA (Sheth et al., 2025) and CodeMixBench (Yang and Chai, 2025). Mechanistically, RomanLens (Saji et al., 2025) finds latent romanization inside multilingual models, and Karne (2026) shows concept representations are not script-invariant using Serbian digraphia. All of this measures the model as a task performer or probes its internals; none measures the model as a judge.

Tokenizer inequity. Tokenizers price languages unequally (Petrov et al., 2023; Ahia et al., 2023), tokenizer choice materially affects downstream performance (Ali et al., 2024), and vocabulary allocation drives quality for low-resource scripts (Liang et al., 2023). For Indic scripts specifically, byte-fallback fragmentation has been quantified as an eight-fold token tax (Srivastava, 2026), with morphology-aware remedies proposed (Limisiewicz et al., 2024; Brahma et al., 2025). This literature stops at token counts and task accuracy; we test, and reject, its natural prediction for judge scores.

Indic evaluation. Benchmarks for Indic languages evaluate in the conventional native script only: MEGA (Ahuja et al., 2023), IndicXTREME (Doddapaneni et al., 2023), IndicGenBench (Singh et al., 2024), MILU (Verma et al., 2025). Airavata (Gala et al., 2024) exemplifies current practice: its Hindi generations are GPT-4-judged in Devanagari, with the judge itself unvalidated. Our result says that practice is sensitive to an unexamined variable.

3 Method

Dataset. We construct 150 Hindi instruction and response pairs across ten everyday domains (science, history, cooking, health, technology, culture, geography, personal finance, daily-life advice, education), at three intended quality tiers of 50 items each: high (correct, complete, 80 to 140 words), medium (correct but visibly incomplete for the ask, 40 to 80 words), and low (curt, vague, or trivially shallow, 15 to 40 words, a few items containing deliberate minor factual errors). Responses were authored in Devanagari and frozen before any judging. Tier intent was strongly realised in the measured scores: Devanagari means of 99.1, 51.4, and 15.2 under Gemini 3.6 Flash.

Conditions. Each frozen response is rendered in four orthographies. *deva* is the original Devanagari. *iast* is scholarly romanization with diacritics, produced deterministically with the indic-transliteration library, in the tradition of transliteration tooling for Indic languages (Madhani et al., 2023). *ascii* strips the diacritics from IAST, a deterministic proxy for plain-keyboard romanization. *hinglish* is a natural user-style respelling (for example *kaise hain aap*), produced under a strict no-paraphrase rule: every word, the word order, and all punctuation preserved. A native-speaker audit of a stratified 20-item sample of the hinglish condition is part of the protocol; the audited sample and its verdict ship with the released data.

Judges and protocol. Six judges from three families score every item in every condition on a 0 to 100 rubric (correctness, completeness, helpfulness, clarity). Gemini 3.6 Flash and Gemini 3.1 Pro Preview are called through the API at temperature 0, one item per call, in randomized order, so the judge never sees two versions of the same item in one context. Claude Sonnet 5, Opus 5, and Fable 5 judge through blind agent sessions arranged in a Latin square: each session sees each item exactly once, in exactly one condition, with conditions rotated across four sessions. Qwen2.5-7B-Instruct runs open-weights under vLLM on one A10G GPU at temperature 0, one item per call, with top-20 logprobs recorded at the score position. No judge is told the study concerns scripts. The judging prompt asks for a score and a one-sentence reason; the reasons become data for the perception analysis (§4.4).

Analysis. The unit of analysis is the within-item paired difference, romanized minus Devanagari, per judge and condition. We report mean shifts with bootstrap 95 percent confidence intervals (10,000 resamples), two-sided sign-flip permutation tests (20,000 permutations), and the paired effect size d_z. All numbers in this paper are produced by one scripted analysis reading the raw score files; the analysis output ships with the paper.

Pre-registration. Before data collection we recorded the predicted direction: romanized scores lower than Devanagari. The observed direction is the opposite. We report all pre-registered analyses regardless.

4 Results

4.1 Three families, three bias profiles

Table 1 and Figure 2 are the study in one view. Gemini 3.6 Flash scores the same content higher when romanized: $+3.43$ points for IAST ($p = 5 \times 10^{-5}$, $d_z = 0.49$), $+2.17$ for ASCII ($p = 0.003$), and $+3.13$ for Hinglish ($p = 5 \times 10^{-5}$, $d_z = 0.52$). Gemini 3.1 Pro shows the same sign at

Qualitative example [item M02, medium tier]. *Same content, one script flip, twenty points.*

Devanagari rendering **score 60/100**

रेफ्रिजरेटर में एक विशेष गैस होती है जिसे रेफ्रिजरेंट कहते हैं। कंप्रेसर इस गैस को दबाकर पाइपों में घुमाता है। जब यह गैस फैलती है तो अंदर की गर्मी सोख लेती है ...

Judge rationale: "The response correctly explains the basic concept, but fails to provide a detailed, step-by-step breakdown ..."

Hinglish rendering (identical content) **score 80/100**

Refrigerator mein ek vishesh gas hoti hai jise refrigerant kehte hain. Compressor is gas ko dabakar pipon mein ghumata hai. Jab yeh gas failti hai to andar ki garmi sokh leti hai ...

Judge rationale: "The response accurately explains the concept *in simple Hinglish*, but lacks a detailed step-by-step breakdown as requested."

Both rationales identify the same deficiency (missing step-by-step detail), and the Hinglish rationale even names the script, yet the identical content scores twenty points higher romanized. Judge: Gemini 3.6 Flash, temperature 0.

Table 1: Mean within-item score shift (romanized minus Devanagari) for every judge and romanization, with bootstrap 95% confidence intervals over $n{=}150$ paired items. Positive means the identical content scored *higher* romanized. **Bold** marks shifts whose permutation $p < 0.01$. The three families disagree in both size and sign: Gemini inflates everywhere, Qwen inflates formal romanizations only, Claude is flat except one small penalty.

Judge	IAST	ASCII	Hinglish
Gemini 3.6 Flash	+**3.43** [+2.4, +4.6]	+**2.17** [+0.8, +3.7]	+**3.13** [+2.2, +4.1]
Gemini 3.1 Pro	+**1.17** [+0.6, +1.8]	+**1.10** [+0.4, +1.8]	+**1.13** [+0.5, +1.8]
Qwen2.5-7B	+**7.63** [+4.9, +10.4]	+**6.03** [+3.3, +8.7]	+0.07 [−1.8, +1.9]
Claude Sonnet 5	+0.11 [−0.7, +0.9]	−0.53 [−1.2, +0.1]	−0.24 [−1.0, +0.5]
Claude Opus 5	+0.02 [−0.5, +0.6]	−**1.23** [−2.0, −0.4]	−0.02 [−0.6, +0.6]
Claude Fable 5	+0.02 [−0.4, +0.4]	−0.54 [−1.2, +0.1]	−0.13 [−0.6, +0.3]
Flash + "ignore script" line	+**2.96** [+1.6, +4.4]	+**3.83** [+1.9, +6.1]	+**2.94** [+1.4, +4.6]

a third the size (all $p \leq 0.003$). The bias is smaller at the stronger tier but it does not vanish.

The open-weights judge changes shape. Qwen2.5-7B inflates formal romanizations dramatically, $+7.63$ for IAST and $+6.03$ for ASCII, but is indistinguishable from zero on natural Hinglish. Its inflation also lives in a different place: low-quality items shift $+17.8$ (IAST) and $+17.4$ (ASCII) points, meaning the small judge fails to recognise bad content once it is dressed in diacritics, while Hinglish, plentiful in web training data (Sheth et al., 2025), is judged as soberly as Devanagari.

The Claude family behaves differently again. Sonnet 5, Opus 5, and Fable 5 shift by at most 0.54 points in eight of their nine judge-by-condition comparisons ($p \geq 0.09$). The single exception is the only significant result in our pre-registered direction anywhere in the study: Opus 5 penalises diacritic-stripped ASCII by -1.23 points ($p = 0.002$), and that is the condition that most visibly degrades the text.

Three conclusions follow. Script bias in LLM judges exists. It is a property of the judge, not of the task: same items, same protocol, three different behaviours across three families. And it is not monotone in romanization naturalness: the invariant family is invariant everywhere, the mid-tier family inflates every romanization moderately, and the small open judge inflates exactly the romanizations it saw least in training.

4.2 The bias lives where the judge has discretion

Figure 3 splits the shift by quality tier. Under Gemini 3.6 Flash, high-quality items shift by at most 1.3 points in either direction against a ceiling near 99. Low-quality items shift by at most 2.7 points against a floor near 15. Medium-quality items, which score near 51 in Devanagari, inflate by $+7.8$ (IAST), $+7.4$ (ASCII), and $+6.5$ (Hinglish) points, all $p \leq 1.5\times10^{-4}$. The same shape holds for Pro at smaller magnitude. This is the pattern one expects if romanization blurs the judge's view of quality: content whose flaws are obvious stays low, content whose merits are obvious stays high, and ambiguous content drifts toward a generous middle. The qualitative example box shows one such drift verbatim: identical refrigerator explanations, the same deficiency named in both rationales, and a twenty-point gap.

4.3 Telling the judge to ignore the script does not work

The remedy every practitioner would try first is one sentence in the judge prompt. We ran the full four-condition protocol again under Gemini 3.6 Flash with the added line "Evaluate the content regardless of the script or orthography it is written in." Figure 4 and the last row of Table 1 show the outcome: medium-tier inflation of $+7.2$ (IAST), $+6.1$ (ASCII), and $+7.2$ (Hinglish) points with the instruc-

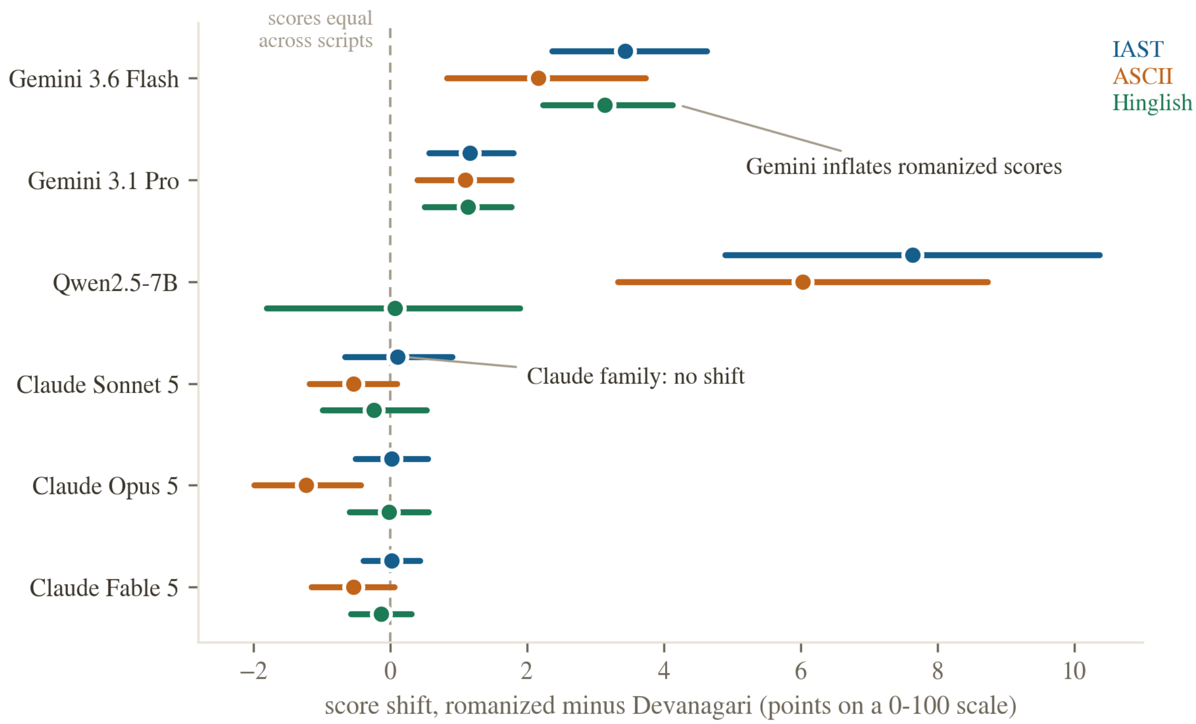

Figure 2: Score shift by judge and romanization, with 95 percent bootstrap confidence intervals. Each row group is one judge; each interval is the mean within-item shift (romanized minus Devanagari) over $n = 150$ paired items. Read intervals against the dashed zero line: right of zero means romanized content scored higher than the same content in Devanagari. Both Gemini judges and Qwen (on formal romanizations) sit right of zero; all three Claude judges straddle zero except Opus 5 on ASCII.

tion, against $+7.8$, $+7.4$, and $+6.5$ without it. The confidence intervals overlap almost completely. The bias is not a surface behaviour the model can suspend on request, which is consistent with debiasing results showing prompt-level fixes underperform training-level ones (Zhou et al., 2026b).

4.4 Mechanism: not token length; the script is perceived

Two probes narrow the explanation space. First, tokenization. A standing hypothesis from the tokenizer-fairness literature is that length disparities drive downstream behaviour (Petrov et al., 2023; Ahia et al., 2023; Srivastava, 2026). We measured every item in every condition with the Gemini tokenizer: romanized Hindi is the *expensive* form, costing 1.98 (IAST), 1.47 (ASCII), and 1.33 (Hinglish) times the tokens of the same content in Devanagari (Figure 5, left). This is itself a finding worth recording, because it reverses the premise of the romanization-efficiency literature, which reported 2 to 4-fold token *savings* from romanization (Husain et al., 2024); modern tokenizers have since closed the Devanagari gap (Brahma et al., 2025). But token inflation does not explain the score inflation: the per-item correlation between token ratio and score shift is $r = +0.11$ (IAST), $+0.05$ (ASCII), and -0.08 (Hinglish), and no tier-restricted correlation exceeds $|r| = 0.18$. Longer input is not what earns the higher score.

Second, perception. The judge's own rationales show it sees what it is scoring (Figure 5, right): under Gemini 3.6 Flash, 47 percent of IAST, 57 percent of ASCII, and 41 percent of Hinglish rationales explicitly mention the script or romanization, against 1 percent for Devanagari. Some rationales even name the orthography as a deficiency while the score lands higher than the Devanagari twin's. The bias therefore operates despite awareness, consistent with the mitigation failure: this is a miscalibration of quality assessment through an unfamiliar surface form, echoing at the behavioural level the finding that internal representations are not script-invariant (Karne, 2026; Saji et al., 2025).

5 Limitations, narrated

We narrate the study's weaknesses as part of the story, because several were design choices made for speed and budget, and a reader should weigh them.

The dataset is model-authored. All 150 items and their quality tiers were generated by a language model under human-specified tier definitions, then frozen. Tier intent was strongly realised in the scores, but model-authored

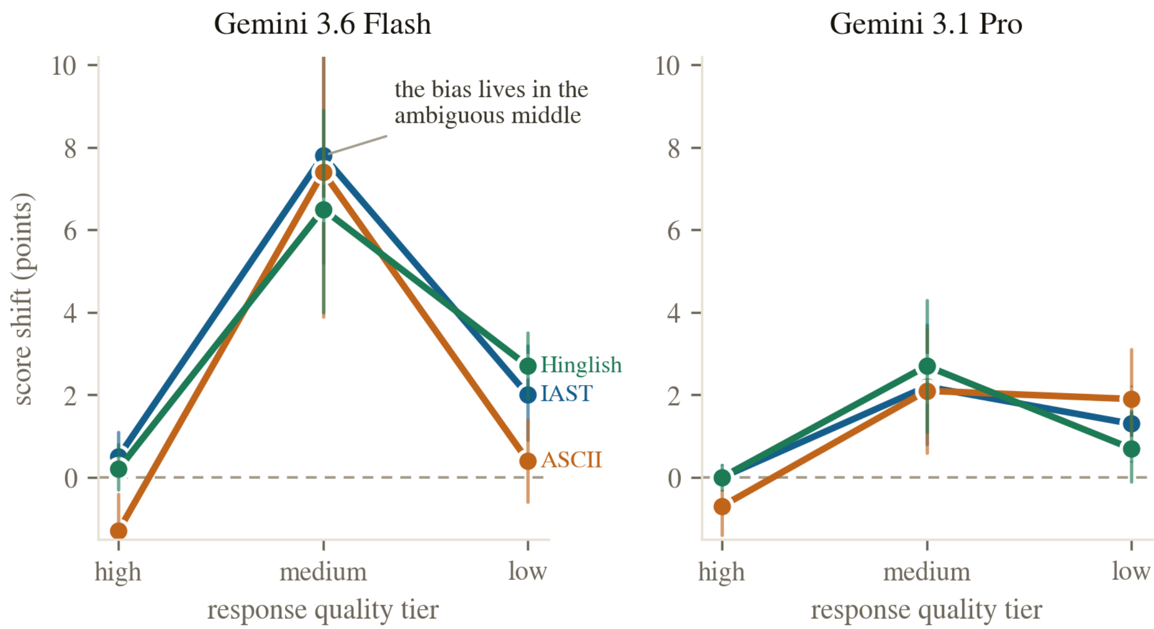

Figure 3: The bias concentrates where the judge has discretion. Mean shift by intended quality tier for the two Gemini judges, with 95 percent bootstrap intervals ($n = 50$ items per tier). High-tier items sit near the score ceiling (Devanagari mean 99.1 under Flash) and low-tier items near the floor (15.2), leaving little room to move; medium-tier items (Devanagari mean 51.4) inflate by 6.5 to 7.8 points under Flash and 2.1 to 2.7 under Pro.

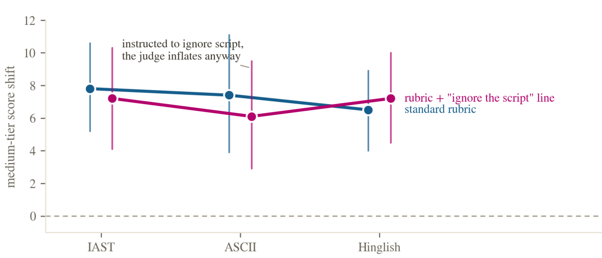

Figure 4: The mitigation fails. Medium-tier mean shift under Gemini 3.6 Flash with the standard rubric versus the same rubric plus the line "Evaluate the content regardless of the script or orthography it is written in." Intervals are 95 percent bootstrap CIs over $n = 50$ medium-tier items. The instructed judge inflates as much as the uninstructed one.

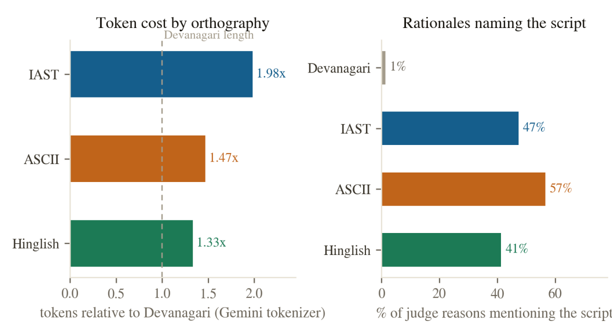

Figure 5: Left: the same content costs 1.33 to 1.98 times as many tokens romanized as in Devanagari under the Gemini tokenizer (mean over 150 items; Devanagari mean 93 tokens). **Right:** share of the judge's one-sentence rationales that mention script, romanization, or transliteration, by condition, under Gemini 3.6 Flash ($n = 150$ per condition). The judge names the script on 41 to 57 percent of romanized items and on 1 percent of Devanagari items.

Hindi may differ from human Hindi in ways that interact with romanization. The hinglish condition is likewise a model respelling under a strict no-paraphrase rule; the protocol includes a 20-item native-speaker fidelity audit, and the audited sample with its verdict ships with the data.

The Claude judges were run through interactive agent sessions, not a version-pinned API, and each session scored 150 items in one context rather than one item per call. The Latin-square rotation keeps the script comparison fair, and no session could compare scripts within an item, but session-based judging is less controlled than the API arms, and items were blocked by tier within a packet, which sessions noticed. One of the three Claude judges shares a model family with the orchestrator of this study; the blind packets mitigate, and the Claude results are null results, which self-favouritism would not produce.

Scale and scope are modest: one language, 150 items, six judges, one seed at temperature 0. The mitigation test covers one instruction phrasing on one judge. Our quality tiers are coarse; a continuous-quality design would resolve the discretion story more finely. Extending the audit to other digraphic settings, such as Serbian Cyrillic and Latin (Karne, 2026) or romanized Sinhala (Rajapakse and Weerasinghe, 2026), is the natural next step.

Finally, the pre-registration was directional and informal (recorded in the project log before data collection), not a third-party registry entry. We state it because it disciplined our reporting: the headline effect is opposite to what we predicted.

6 Conclusion

We held Hindi content fixed, changed only its writing system, and asked six LLM judges to grade it. The writing system moved the grades. Gemini judges inflate romanized Hindi by 2 to 3 points overall and 7 to 8 points on ambiguous-quality content; the effect shrinks threefold but survives at the frontier tier; a 7B open-weights judge inflates formal romanizations by up to 18 points on low-quality content while judging natural Hinglish soberly; Claude judges are script-invariant except for a small penalty on the most degraded romanization; a script-blindness instruction changes nothing; token length does not explain it; and the judges' own rationales prove the script is seen. The bias axis is real, it is judge-dependent in both size and sign, and it belongs in every judge-bias taxonomy (Ye et al., 2025; Zhou et al., 2026b) and every multilingual evaluation checklist (Hada et al., 2024b; Son et al., 2024). Until judges are audited for script invariance, evaluations of Hindi systems should report native-script and romanized slices separately, and should say which judge produced the numbers.

Reproducibility

All datasets, per-call score files, the scripted analysis, figure build scripts, and spend logs are released at `https://github.com/Abraar237/script-bias-llm-judges`, with a narrative project page at `https://abraar237.github.io/script-bias-llm-judges/`. Every number above traces to `results/analysis.json`, generated by `experiments/analyze.py` from the raw judgment files. Total marginal compute cost of the study: under 5 US dollars of API spend plus roughly one A10G GPU-hour.

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